DAM501 – DATA MINING

Final Project: Knowledge Discovery from Data for Decision-Making

This final project focuses on the application of Data Mining techniques to discover meaningful patterns, structures, and actionable knowledge from real-world data in a business context. It emphasizes the knowledge discovery process, including data understanding, pattern exploration, insight generation, and interpretation for decision-making.

Students are expected to apply the Data Mining pipeline, from business understanding and data preparation to pattern mining, model interpretation, and business insights.

1. Problem Definition and Business Understanding (15%)

Provide background information about the business domain or industry related to your dataset (e.g., retail, finance, healthcare, e-commerce, telecommunications).

Clearly describe the **business challenge or opportunity** that can be addressed through data mining. The focus should be on understanding behavior, discovering structure, or uncovering hidden patterns, rather than only predicting outcomes.

Example:
Understanding customer purchasing behavior to identify common product combinations and customer segments for targeted marketing.

Data Mining Objective

Clearly define the **data mining task(s)** involved in your project. You may include one or more of the following:

- Clustering (e.g., customer or product segmentation)
- Association Rule Mining (e.g., market basket analysis)
- Sequential Pattern Mining
- Anomaly or Outlier Detection
- Descriptive or Exploratory Pattern Discovery
- Classification or Regression (supports insight discovery)

Justify **why these data mining tasks are appropriate** for the business problem.

Business Value

Explain how the discovered patterns or insights can support business decisions, such as:

- Improving marketing strategies

- Enhancing customer retention
- Optimizing product placement
- Detecting abnormal or risky behavior

Assumptions and Limitations

List any assumptions made about the data or business context, and discuss limitations such as data quality, scope, or generalizability of findings.

2. Dataset Selection and Understanding (5%)

- Select a **real-world dataset** relevant to your business problem.
- The dataset may be structured or transactional data.
- Public datasets can be obtained from sources such as Kaggle, UCI Machine Learning Repository, government portals, or open business datasets.
- Clearly describe:
    - Data source
    - Size of the dataset
    - Types of attributes
    - Data collection context (if available)

3. Data Pre-processing and Transformation (10%)

Demonstrate appropriate data preparation techniques to improve data quality and support effective data mining. This may include:

- Handling missing values
- Detecting and treating outliers
- Data cleaning and noise reduction
- Feature transformation or discretization
- Data aggregation or transaction construction
- Encoding categorical variables

Explain **why each step is necessary** and how it affects the mining results.

4. Exploratory Data Analysis and Descriptive Mining (15%)

Conduct an in-depth exploratory analysis to **understand the structure and characteristics of the data**.

This section should focus on **discovering patterns and relationships**, not just preparing data for modeling. Possible analyses include:

- Distribution analysis and summary statistics
- Group-based comparisons
- Behavioral or usage pattern exploration
- Preliminary clustering or visualization
- Identification of unusual or interesting trends

Summarize key findings that guide the choice of data mining techniques.

5. Data Mining Methods and Pattern Discovery (30%)

Apply **at least two different data mining techniques**, with **at least one unsupervised or pattern-based method**, such as:

- Clustering (e.g., K-Means, Hierarchical Clustering, DBSCAN)
- Association Rule Mining (e.g., Apriori, FP-Growth)
- Sequential Pattern Mining
- Anomaly Detection
- Classification or Regression

For each method:

- Explain why the method was chosen
- Describe the core idea of the algorithm at a conceptual level
- Explain parameter selection and tuning
- Present the discovered patterns, clusters, or rules

The emphasis should be on **interpretation and insight**, not algorithmic complexity.

6. Evaluation and Interpretation of Results (15%)

Evaluate the quality and usefulness of the discovered patterns using appropriate criteria, such as:

- Interpretability and clarity
- Support, confidence, and lift (for association rules)
- Cohesion and separation (for clustering)
- Stability and consistency of patterns
- Business relevance and actionability

Discuss the **strengths and weaknesses** of each data mining approach in terms of:

- Insight quality
- Scalability
- Interpretability

- Practical usefulness

7. Business Insights and Recommendations (10%)

Translate the data mining results into actionable business insights.

- What did the business learn from the data?
- Which patterns are most valuable?
- How can these insights support real-world decisions?

Provide concrete recommendations, such as:

- Customer segmentation strategies
- Product bundling or cross-selling opportunities
- Risk or anomaly monitoring mechanisms
- Policy or operational improvements

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8. (Optional Bonus) Advanced Data Mining Techniques (+10%)

Apply advanced techniques to enhance insight discovery, such as:

- Feature engineering for better pattern discovery
- Dimensionality reduction for structure exploration
- Novel mining approaches yieding competitive or SOTA results
- Comparative analysis across different subsets of data

Discuss how these techniques improve understanding rather than just performance.

Deliverables

1. Project Report (PDF or Word)

The report should be clearly structured and include:

1. Introduction and Business Understanding
2. Dataset Description
3. Data Pre-processing
4. Exploratory and Descriptive Analysis
5. Data Mining Methods and Results
6. Evaluation and Interpretation
7. Business Insights and Recommendations

Include tables, figures, and visualizations where appropriate.

2. Code Submission

- Clean, well-documented, and reproducible code
- Clearly separate data preparation, mining, and analysis steps

3. Presentation

Prepare a short presentation summarizing:

- Business problem
- Key patterns and insights discovered
- Data mining techniques used
- Business implications